

Sequence analysis

Predicting disulfide bond connectivity in proteins by correlated mutations analysis

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ABSTRACT

Motivation: Prediction of disulfide bond connectivity facilitates structural and functional annotation of proteins. Previous studies suggest that cysteines of a disulfide bond mutate in a correlated manner.

Results: We developed a method that analyzes correlated mutation patterns in multiple sequence alignments in order to predict disulfide bond connectivity. Proteins with known experimental structures and varying numbers of disulfide bonds, and that spanned various evolutionary distances, were aligned. We observed frequent variation of disulfide bond connectivity within members of the same protein families, and it was also observed that in 99% of the cases, cysteine pairs forming non-conserved disulfide bonds mutated in concert. Our data support the notion that substitution of a cysteine in a disulfide bond prompts the substitution of its cysteine partner and that oxidized cysteines appear in pairs. The method we developed predicts disulfide bond connectivity patterns with accuracies of 73, 69 and 61% for proteins with two, three and four disulfide bonds, respectively.

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1 INTRODUCTION

The disulfide bond is the most frequent naturally occurring covalent cross-link in proteins. It is derived from the oxidation of the thiol groups of two cysteine residues. Proteins with disulfide bonds are usually secreted and rarely found in the cytoplasm, which has a reducing environment and lacks enzymes that promote disulfide bond formation (Kadokura *et al.*, 2003). However certain archaea are rich in cytoplasmic proteins with disulfide bonds (Mallick *et al.*, 2002). A number of studies have linked disulfide bonds to protein stability and to folding rate (Wedemeyer *et al.*, 2000). It has been suggested that disulfide bonds stabilize the protein's folded state by restricting the protein's conformation, thereby reducing the entropy of the unfolded state (Harrison and Sternberg, 1994; Poland and Scheraga, 1965). Meanwhile, disulfide bonds increase the enthalpy of the folded state by stabilizing local interactions (Wedemeyer *et al.*, 2000). Furthermore, disulfide bonds increase the protein's half-life by enhancing protein protection

against proteases by maintaining the integrity of protein structure against local unfolding events. Disulfide bonds have also been observed to contribute to protein function regulation (Hogg, 2003).

Disulfide bonds constrain the conformation of the protein structure and thus, knowledge of their location can facilitate protein structure prediction. In addition, disulfide bond connectivity patterns can be used to discriminate between protein folds and to accurately superimpose protein structures (Chuang *et al.*, 2003; Gupta *et al.*, 2004; Mas *et al.*, 1998). The underlying assumption in these methods is that similar disulfide bond connectivity patterns place similar spatial constraints on proteins, resulting in similar protein structures. Finally, variation in disulfide bridge patterns may be used to infer variation of protein function (Cao *et al.*, 2007).

There are two distinct steps in the process of predicting disulfide bond connectivity patterns. The first is the classification of bound (oxidized) and unbound (reduced) cysteines. The second is the correct pairing of all bound cysteines. Muskall and colleagues (1990) published the first method to identify bound and free cysteines by utilizing a neural network and reported a prediction accuracy of 82%. Fiser and colleagues (1992) observed that the sequence environments of bound and free cysteines have different compositions and they subsequently introduced a method to calculate disulfide-bond forming potential that is based on the amino acid composition of the sequential environment of cysteines. Later, Fiser and Simon analyzed the apparent difference between the conservation level of oxidized and reduced cysteines and developed a method to predict the oxidation states of cysteines from the conservation analysis of multiple sequence alignments. This simple approach reached a prediction accuracy of 82% (Fiser and Simon, 2000). In the same study it was noted that it is rare for a protein to have cysteines with mixed oxidation states. Mucchielli-Giorgi *et al.* (2002) developed a cysteine oxidation state predictor based on Fiser and Simon's finding (Fiser and Simon, 2000) and on the global amino acid composition of proteins and attained an 84% prediction accuracy. Chen *et al.* (2004) trained a support vector machine on the local environment of cysteines as well as on global information of the protein and reported a 90% prediction accuracy. Given these high prediction accuracies for cysteines oxidation state prediction our current study focuses on the second step in prediction of disulfide bond

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connectivity; the challenging problem of identifying the correct pairing of bound cysteines.

Given $2n$ cysteines that form n disulfide bonds, the number of possible connectivity patterns of all $2n$ cysteines is $(2n-1)!!$. The number of possible disulfide bond connectivity patterns (cysteine pairings) increases rapidly with the number of bound cysteines (e.g. for proteins with four, six and eight disulfide bonds there are 105, 10 395 and $\sim 2 \times 10^6$ possible connectivity patterns, respectively). An exhaustive search for optimal pairing of cysteines is possible only when the number of bound cysteines is small. To overcome the combinatorial explosion problem, the problem of pairing bound cysteines was translated into the problem of finding the perfect match in a complete weighted and undirected graph (Fariselli and Casadio, 2001), which can be solved in polynomial time using the Edmund-Gabow algorithm (Gabow, 1976). In their approach, graph vertices, edges and the weights of edges represent bound cysteines, potential connectivity between two cysteines and confidence scores for the pairing of two cysteines, respectively (Fariselli and Casadio, 2001).

Most current methods that predict disulfide bond connectivity use graph representation. These methods typically differ in the way in which the weights of the edges are calculated. Fariselli and Casadio (2001) assigned contact potentials to edge weights based on the assumption that the nearest sequential neighbors of the paired cysteines were also in contact. Their calculation was limited to protein queries with up to five disulfide bridges as the process of calculating the contact potential employed time consuming Monte Carlo and simulated annealing procedures. In a more recent work, the authors increased the speed of the contact potential calculation by employing a neural network (Fariselli *et al.*, 2002). Vullo and Frasconi (2004) were able to significantly increase the accuracy of prediction by incorporating evolutionary information. The authors utilized a recursive neural network to score disulfide connectivity patterns. Ferre and Clote (2005) utilized a neural network with a unique hidden layer intended to examine bi-residue information. They incorporated evolutionary information in the form of Position Specific Scoring Matrices (PSSM) and also added secondary structure information. Tsai *et al.* (2005) used a Support Vector Machine (SVM) with evolutionary information and protein sequence separation of cysteines pairs as inputs. Cheng *et al.* (2006) created a complete platform for disulfide bridge prediction by predicting both the bound state of the cysteines and the disulfide bond connectivity pattern. The authors utilized kernel methods to predict the bound state of cysteines and a recursive neural network to predict disulfide bond pairing. The input to the neural network included evolutionary information, sequence separation of cysteine pairs, and solvent accessibility. Zhao *et al.* (2005) approached the problem of identifying the correct pairing of bound cysteines from a global perspective. Rather than scoring each possible pair of cysteines, the authors compared the query cysteine sequence separation profile to a database of similar profiles of proteins with known disulfide bonds. The limitation of the approach is that a novel cysteine pattern cannot be found. Chen and Hwang (2005) incorporated both local evolutionary information, in the form of PSSM, and global information, in the form of cysteine separation profile, as inputs for a SVM.

Lu *et al.* (2007) used a genetic algorithm to improve the optimal selection of input sources for disulfide bond prediction. Chen *et al.* (2006) utilized global and local information as inputs for a two layer SVM. The first layer had a SVM that utilized local information as the input. The inputs for the second layer were scores from the first layer of SVM along with global information such as the protein length, the cysteine separation profile and the disulfide connectivity frequency.

Fiser and Simon (2000) showed that it is possible to discriminate between the different oxidation states of cysteines based solely on the conservation analysis of the cysteines. However, this analysis cannot predict the correct pairing of oxidized cysteines because all the oxidized cysteines are expected to have a similar level of conservation (Fiser and Simon, 2002). Two cysteine residues in a disulfide bond form a strong interaction, which, in many cases, maintains both the protein's structure and function. Such a strong interaction is expected to lead to interdependency between the two positions, which could be traced through evolution. In addition, it is difficult to maintain different redox conditions for the same protein environment, i.e. if a bridge forming an oxidized cysteine is mutated, its reduced cysteine partner will be under pressure to mutate as well. Correlated mutation algorithms aim to identify residue-residue linkages through identifying patterns of concerted variations in different positions in a multiple sequence alignment. A variety of correlated mutation algorithms have already been utilized for predicting residue contacts in a protein 3D structure (Dekker *et al.*, 2004; Gobel *et al.*, 1994; Hamilton *et al.*, 2004; Larson *et al.*, 2000; Neher, 1994; Shindyalov *et al.*, 1994), but there has not yet been an attempt to utilize correlated mutation algorithms to automatically identify disulfide bond connectivity.

Thornton (1981) analyzed 15 cases of non-conserved disulfide bonds and observed that when a disulfide bond is not conserved both cysteines are mutated in concert. Kreisberg *et al.* (1995) examined the trypsin-like serine proteases and phospholipase A2 protein families and demonstrated that cysteines forming disulfide bonds mutate in a correlated pattern. The authors noted that this correlated pattern could be used to predict disulfide bonds in proteins. These two studies observed trends in cysteine mutations with respect to conservation of disulfide bonds, although they utilized very small databases.

This study uses a large set of protein families to analyze non-conserved disulfide bonds. Subsequently, we introduce a novel method that predicts disulfide bond connectivity pattern using a correlated mutation algorithm.

2 METHODS

2.1 Conservation analysis of disulfide bonds

2.1.1 Data set In order to assess the conservation pattern of disulfide bonds, we examined protein domains from (SCOP) structural classification of proteins (Murzin *et al.*, 1995) that had at least one disulfide bond. Our operational definition for a disulfide bond occurrence is when Sulfur gamma (SG) atoms of two cysteine residues fall within a 2.5 Å distance of each other. The expected SG-SG distance for disulfide bond is ~ 2 Å but this more generous definition accounts for inaccuracies in experimental data. We removed redundancy of

proteins at a 90% sequence identity level using CD-HIT (Li and Godzik, 2006).

2.1.2 Generating multiple structural alignments Multiple structural alignments of each SCOP family were generated with multiple structural alignment algorithm (MUSTANG) (Konagurthu *et al.*, 2006). Since MUSTANG aims to optimize alignment of all residues in the proteins, yet we were looking specifically for optimal alignments of disulfide bonds, we realigned the cysteine pairs for disulfide bonds that were found to be misaligned. A disulfide was assumed to be misaligned if cysteines were found within five alignment positions off of a common disulfide bond position.

2.2 Disulfide bond prediction

2.2.1 Data set In order to benchmark the predictive power of our method we used the same version of annotated protein sequences of Swiss-Prot as other studies: release 39 (1999) (Boeckmann *et al.*, 2003). Protein sequences were filtered by two requirements as described previously (Fariselli and Casadio, 2001). First, only proteins with known 3D structures were considered. Second, disulfide bonds annotation could not contain the words ‘by similarity’, ‘probable’ or ‘potential’. The test set had 435 proteins that were grouped by the number of disulfide bonds.

2.2.2 Generating multiple sequence alignment For each query protein, evolutionary related sequences were extracted from NR (Wheeler *et al.*, 2007) by running five rounds of PSI-BLAST (Altschul *et al.*, 1997). A representative multiple sequence alignment was generated by filtering the sequences from the PSI-BLAST output using BlastProfiler (Rai *et al.*, 2006) with the following parameters: minimum *e*-value lower than 0.0001, hit-query alignment sequence identity of at least 15%, hit-query alignment coverage of at least 30% and 90% maximum sequence identity between any two hits.

2.2.3 Scoring scheme Our scoring scheme is based on the correlated substitution pattern observed for positions participating in disulfide bonds. In most cases of non-conserved disulfides both cysteines are substituted (see Section 3). Since it is unlikely that all disulfide bonds will always mutate simultaneously, we search for a simple correlation pattern of concerted appearing and disappearing of cysteines in order to predict disulfide bonds. Given a multiple sequence alignment, we examined only those sequence positions (columns) that correspond to disulfide forming cysteine positions in the query. For each sequence in the alignment, we divided the examined positions into two sets based on their amino acids composition; the first set is composed of positions with cysteine residues, while the second set is composed of positions with a gap or any residue other than a cysteine. The score for each possible pair in a sequence is a number between zero and one, and corresponds to our expectation that this pair of positions form a disulfide bond in the query protein based solely on the current sequence examined. If two positions are part of different sets (only one position is a cysteine) then the correlation score is zero, because our observation demonstrated that it is unlikely that only one of the positions that formed a disulfide bond in the query is substituted. If two positions are part of the same set (either a set of all cysteines or a set of anything but cysteines) then the score for a pair of such positions is $1/(size\ of\ set - 1)$, which is the probability of selecting the correct position pair randomly and with equal chance, assuming that pairing is possible between only those positions that are part of the same set. Those sequences were ignored in the alignment that contained either completely conserved or completely varied all cysteines as no correlated mutation information can be extracted from these. Also, sequences with an odd number of cysteines, at the sequence positions examined, were removed as they were assumed to be a product of a misalignment.

Scores for pairing all possible combinations of all positions for each of the aligned sequences were collected in a matrix that represented all possible disulfide bond combinations in the query. Averaging the scores in all matrices generate a final scoring matrix. Next, we exhaustively generated all possible disulfide bond connectivity patterns and scored them by summing up the scores of the individual disulfide bonds using values from the final scoring matrix. A global score is reported for each possible disulfide bond connectivity pattern.

The steps described are formalized below:

$A = \{\text{Alignment positions (columns) corresponding to the query bound cysteines}\}$

$I^j = \{i \mid i \in A \ \& \ \text{Position } i \text{ in sequence } j \text{ is a cysteine}\}$

$II^j = \{i \mid i \in A \ \& \ \text{Position } i \text{ in sequence } j \text{ is not a cysteine}\}$

$$M_{n,m}^j = \begin{cases} 0, n \in I^j \ \& \ m \in II^j \\ 0, n \in II^j \ \& \ m \in I^j \\ \frac{1}{|I^j|-1}, n, m \in I^j \\ \frac{1}{|II^j|-1}, n, m \in II^j \end{cases}$$

$$M_{n,m} = \frac{\sum_j^N M_{n,m}^j}{N}$$

where N is the number of sequences in the alignment.

2.2.4 Retrospective prediction of disulfide connectivity for year 1999 Multiple sequence alignments were generated as described earlier. Then sequences were removed if, in the NCBI protein flat-file, the description of the year of creation was later than 1999.

3 RESULTS

3.1 Disulfide conservation analysis

3.1.1 Cysteines of disulfide bonds mutate in concert In order to analyze the conservation of disulfide bonds, we constructed multiple structural alignments of proteins of SCOP families that contained at least one disulfide bond. From 189 families 1363 such proteins were analyzed. We examined whether cysteines of non-conserved disulfides mutate in concert by comparing the number of times disulfides were substituted by two non-cysteines residues (or gaps) to the number of times only one of the cysteines was substituted. Disulfide bonds were observed to vary in 4288 cases and in 3463 (81%) of these cases both cysteines mutated in concert. However, after manual investigation it turned out that, in most of the cases, the non-concerted disulfide substitution was a consequence of either a misalignment or a protein structural divergence that the alignment program could not account for. Upon removing such disulfides from our conservation analysis 99% of the cases showed disulfides that mutated in concert.

3.1.2 Disulfide bond forming cysteines are not always conserved Proteins of the same SCOP family have obvious evolutionary relationships, usually sharing >35% sequence identity. Nevertheless, we observed that, even within the same SCOP family, disulfide bonds are not always conserved. Of the families, 60% had at least one non-conserved disulfide bond and, overall, we observed that 66% of all disulfide bonds are not conserved. These results are in agreement with recent findings, which show that variations in the number of disulfide bonds in proteins of the same structural family are not unusual (Cheek *et al.*, 2006).

Table 1. Summary of the accuracy [TP/(TP+FP)] and coverage (predicted queries/all queries) of disulfide connectivity predictions for proteins with two, three and four disulfides

Number of disulfides (number of query proteins)	Current 2007 NR		1999 NR
	Accuracy	Coverage	Coverage
2 (148)	0.73	0.68	0.56
3 (146)	0.69	0.33	0.14
4 (98)	0.61	0.18	0.11

Current NR and 1999 NR—sequences for the multiple sequence alignment are collected from the current (2007) or from the 1999 non redundant protein sequence database, respectively.

Table 2. Accuracy and coverage of *disulfide-bond* predictions for proteins with two, three, four and five disulfide-bonds

Number of disulfides (number of query bridges)	Current 2007 NR		1999 NR
	Accuracy	Coverage	Coverage
2 (296)	0.73	0.68	0.56
3 (438)	0.71	0.43	0.24
4 (392)	0.64	0.34	0.27
5 (215)	0.59	0.43	0.29

3.2 Prediction of disulfide bond connectivity pattern

3.2.1 Performance of the prediction method Our method can predict disulfide bonds in proteins with any number of bonds. However, results are reported only for proteins with 2–4 disulfide bonds because the prediction of proteins with one bond is trivial and sequence databases lack a sufficient number of protein sequences with five or more disulfide bonds for a statistically significant analysis.

Tables 1 and 2 summarize our results for predicting disulfide bond connectivity patterns and disulfide bonds, respectively. When reporting the prediction accuracy of disulfide bond connectivity patterns, we assess our success in predicting the entire disulfide bond connectivity pattern in the protein correctly. In contrast, when reporting on disulfide bonds prediction we measure our ability to correctly predict any disulfide bond in a protein. Tables 3 and 4 list results of predictions obtained in previous studies as well as results from two trivial predictors: a random predictor and a frequency predictor. A random predictor predicts disulfide connectivity by random, while a frequency predictor predicts bridges by relying on the most common connectivity pattern observed in the database. Comparing our results to the random and frequency predictors demonstrates that correlated mutations can capture the evolutionary signal generated by the disulfide bond interactions. The strength of using a correlated mutation analysis is most apparent when predicting connectivity patterns for proteins with four disulfide bonds (105 possible ways to combine four bonds). The method presented here is capable of predicting all four disulfide bonds with 61% accuracy and can

Table 3. Accuracy of predicting disulfide bond connectivity by other methods

Methods	Number of disulfide bonds		
	2	3	4
Random	0.33	0.07	0.01
Frequency ^a	0.58	0.29	0.01
FC ^b	0.56	0.21	0.17
VF ^a	0.73	0.41	0.24
CSB ^c	0.74	0.51	0.27
FCL ^d	0.62	0.4	0.55
CSP ^e	0.74	0.44	0.26
TCCLK ^f	0.79	0.53	0.55
CH ^g	0.74	0.61	0.3
CTCK ^h	0.85	0.67	0.57
GASVM ⁱ	0.86	0.75	0.63

Results are taken from previous works and are reported for proteins with two, three and four disulfide bonds.

References for methods in Tables 3 and 4: (a) (Vullo and Frasconi, 2004), (b) (Fariselli and Casadio, 2001), (c) (Cheng *et al.*, 2006), (d) (Ferre and Clote, 2005), (e) (Zhao *et al.*, 2005), (f) (Tsai *et al.*, 2005), (g) (Chen and Hwang, 2005), (h) (Chen *et al.*, 2006) and (i) (Lu *et al.*, 2007).

Table 4. Accuracy of disulfide bond predictions by other methods

Methods	Number of disulfide bonds			
	2	3	4	5
Random	0.33	0.2	0.14	0.11
Frequency ^a	0.58	0.37	0.1	0.23
FC ^b	0.56	0.36	0.37	0.21
VF ^a	0.73	0.51	0.37	0.30
CSB ^c	0.74	0.61	0.44	0.41
CSP ^d	0.74	0.53	0.44	0.31
TCCHK ^e	0.79	0.62	0.70	0.71
CH ^f	0.74	0.69	0.4	0.31
GASVM ^g	0.86	0.80	0.77	0.71

Results are cited from previous works and reported for proteins with two, three, four and five bonds.

predict a subset of bonds out of the four bridges with an accuracy of 64%. Because, we predicted a subset of the data set utilized by other studies any direct comparison is limited. Nevertheless, in order to obtain a general insight, we evaluated our results along with the results obtained from other methods. With the exception of one method (Lu *et al.*, 2007), our approach produces predictions with the highest accuracies for proteins with three and four disulfide bonds both in predicting disulfide bond connectivity and in predicting disulfide bonds.

3.2.2 Predicting disulfide bonds of proteins with a mixed state of cysteines We also analyzed our prediction method using protein sequences with disulfide bonds but with an odd number of cysteines. In addition to providing information on the predicted disulfide pattern, we also identified the unbound cysteine.

The sequences we analyzed rarely had cysteines with mixed oxidation states in agreement with observations of earlier studies (Fiser and Simon, 2000). Out of the 137 cases, we studied only 12 could be confirmed to have both oxidized and reduced cysteine, for 38 it was not possible to identify the origin of the extra cysteine (not even after consulting the original literature) and 87 cases came from separating cysteines into intra and inter domain disulfide bonds. In these cases the cysteine of the interdomain disulfide bond shows as an unbound one because the crystal structure presents the monomeric state only. This latter set presents a more difficult task to the prediction algorithm, as the conservation levels of intra and inter disulfide bond forming cysteines are rather similar and these differ in their correlation pattern only. Our results for proteins with three, five and seven cysteines (i.e. proteins with one, two and three disulfide bonds and one free cysteine, respectively) demonstrate 91, 55 and 24% prediction accuracy, respectively. In terms of possible number of connectivity combinations, these numbers can be compared to the prediction accuracies of 73, 69 and 61% of the connectivity patterns for four, six and eight cysteines, respectively, when all cysteines are known to be in disulfide bonds. This suggests that with an increasing number of combinations the accuracy of prediction is getting worse, possibly due to an extra task of identifying free/interdomain cysteines. However, the accuracies remain significant and comparable to our earlier results.

3.2.3 Applicability of the method Prediction of the disulfide bond's connectivity pattern using the correlated mutation algorithm presented here requires that all but one disulfide bond is not fully conserved. Our method cannot predict disulfide connectivity patterns of proteins that do not follow this requirement. We evaluated the applicability of our algorithm by measuring the coverage (the number of predicted proteins divided by the number of proteins tested, or the number of disulfide bond predictions divided by the number of disulfide bonds tested) (Tables 1 and 2). The number of predicted proteins with 2–4 disulfide bonds was high enough for a statistically significance analysis. However, there were only nine predicted proteins with five disulfide bonds, which limited the reliability of statistical analysis, and therefore, we did not analyze the accuracy of disulfide bond connectivity predictions for proteins with five or more disulfide bonds. When we tested the performance of the current algorithm using a sequence database from 1999, we found a considerable decrease in coverage, sometimes by half (Tables 1 and 2). This implies that our algorithm applicability will further increase in the future, as sequence databases expand.

3.2.4 Illustration of the prediction method The first example suggests that, in order to accurately predict all disulfide bonds in a query protein, our algorithm may require only a very few evolutionary related sequences to the query as long as they are sufficiently diverse in their disulfide bond patterns. Pepsin-A precursor (pepa_human) is a human protein that belongs to the peptidase A1 family. This protein has three disulfide bonds located at the query sequence positions of 107–112, 268–272 and 311–344. We automatically generated a multiple sequence alignment with 152 sequences but only two of the 152 sequences could be used for correlated mutation analysis because, in the

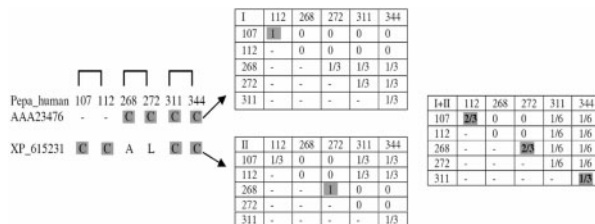


Fig. 1. Pepsin-A precursor (pepa_human) has three disulfide bonds 107–112, 268–272, and 311–344. The alignment positions of pepa_human disulfides with two protein sequences with non-conserved disulfides, and the correlated mutation score matrices corresponding to each sequence is shown. Matrix I+II is the final correlated mutation score matrix, which is obtained by summing and normalizing the two sequence specific scoring matrices (I, and II). The correlation scores of two cysteines that allow unambiguous pairing are highlighted in the matrices.

rest of the cases, all three disulfide bonds were completely conserved. However, the alignment with the two remaining protein sequences (AAA23476, and XP_61523) was sufficient to decipher the connectivity of all three disulfide bonds in the query protein. Each protein sequence in the alignment had a different non-conserved disulfide bond, which is the minimum required information to predict disulfide patterns properly in a protein with three disulfide bonds (Fig. 1).

A second example highlights the correlated mutation pattern observed in disulfide bond positions. β -lactamase (hcbB) from *Helicobacter pylori* has four disulfide bonds formed between cysteines at sequence positions of 22–30, 52–60, 88–96 and 124–132. Figure 2 illustrates the correlated mutation pattern observed in a multiple sequence alignment of hcbB proteins. The corresponding correlated mutation matrix scores generated by our algorithm highlight the simultaneous mutations of both cysteines in non-conserved disulfide bonds.

A third example illustrates the fact that a correlated mutation signal can drastically reduce the problem of disulfide connectivity prediction even if the prediction is partially ambiguous. Proproteinase E precursor (Cac3_bovine) is a bovine protein with 10 cysteines involved in five disulfide bonds located at query sequence positions 41–57, 100–103, 140–206, 171–187 and 196–227. A multiple sequence alignment reveals that two of the five disulfide-bonds are completely conserved (171–187 and 196–227). However, three disulfide bonds can be accurately predicted based on their correlated pattern of conservation. Although our algorithm cannot fully predict all the disulfide bonds of Cac3_bovine, it produces valuable information as it reduces the complexity of prediction from 945 possible combinations of 10 cysteines to three possible combinations of four cysteines. The multiple sequence alignment of Cac3_bovine is composed of 171 sequences of which nine are completely conserved and 16 have an odd number of cysteines at the examined positions (these were ignored as assumed to be product of misalignment). Out of the remaining 146 sequences 131 had one unconserved disulfide bond, which was always aligned with the query disulfide bond at sequence positions 100–103. Fifteen sequences had two unconserved disulfide bonds, of which 13 had neither the disulfides corresponding to the query sequence positions 100–103, nor to the sequence positions 140–206. Two sequences did not have

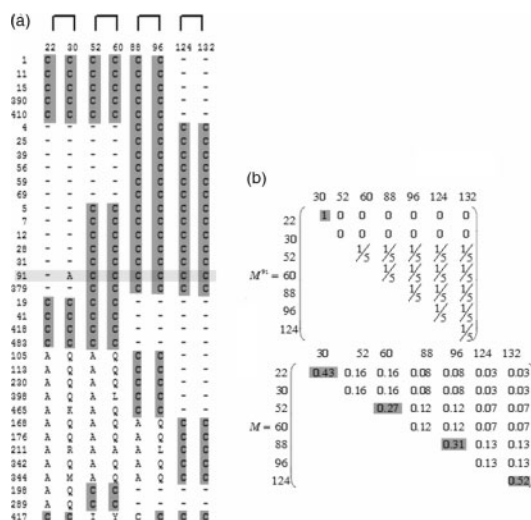


Fig. 2. (a) Columns of a multiple sequence alignment corresponding to bound cysteine positions of β -lactamase (hcpB) protein are shown. The horizontal numbers on the top are the sequence positions of eight cysteines in the hcpB protein that form four disulfide bonds (connectivity pattern is illustrated above). (b) The resulting scoring matrices when applying our correlated mutation algorithm (Section 2) for sequence 91 in the multiple sequence alignment and for all sequences are M^{91} and M , respectively. Based on the M^{91} matrix a prediction for only one bond is possible (22–30). In order to predict the other disulfide bonds at least two other sequences are required (e.g. sequences one and four). Highlighted correlation scores in the matrices allow unambiguous pairing of cysteines.

disulfide bonds corresponding to the query disulfide bonds at sequence positions 41–57 and 100–103 (Fig. 3).

4 DISCUSSION

In the current study, we developed a correlated mutation algorithm to identify disulfide bonds in proteins using sequence information alone. We assumed that correlated mutation analysis is a suitable technique to predict disulfide bonds, because disulfide bond is a well defined residue–residue interaction that plays an important role for the protein structure and function. When such a strong relationship exists between two sequence positions it is expected to result in the coevolution of these positions.

Two requirements have to be fulfilled in order to predict disulfide bonds with correlated mutation analysis. First, some disulfide bonds must be unconserved and second, cysteines of unconserved disulfide bonds have to substitute in a correlation manner. Our analysis of multiple structure alignments of proteins from the same SCOP family demonstrated that both conditions are met. In agreement with recent findings (Cheek *et al.*, 2006), we observed that the number of disulfide bonds varied between evolutionary related proteins. We also demonstrated that multiple sequence alignment columns corresponding to the query disulfide bonds showed a correlated pattern of conservation, i.e. the simultaneous appearance and disappearance of cysteines.

In the current analysis we assumed that the cysteines participating in a disulfide bond are known and, therefore we

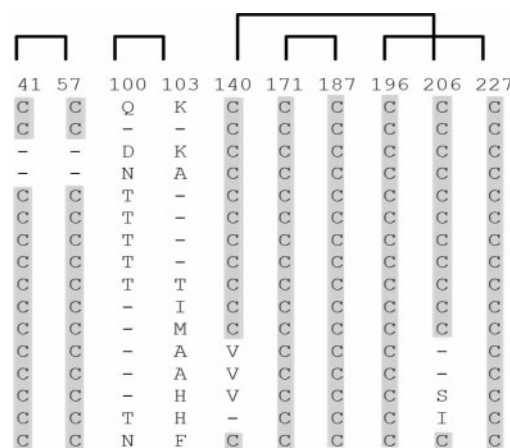


Fig. 3. Columns of a multiple sequence alignment that correspond to oxidized cysteine sequence positions of proproteinase E precursor (Cac3_bovine). Sequence positions that form disulfide bonds are shown above. Alignments positions are shown for a representative subset of sequences that are related to Cac3_bovine but have at least one non-conserved disulfide bond. Conserved cysteines are highlighted.

focused on identifying the correct pairing of these residues. The reason behind this feasible assumption is twofold: (i) the bound state of cysteines can be predicted by several methods with a high accuracy of around 90% (see Section 1), and (ii) <5% of proteins contain cysteines with mixed oxidation states (Fiser and Simon, 2000). Furthermore, past findings have shown that unbound cysteines are significantly less conserved than bound one (Fiser and Simon, 2000), and consequently these will have little or no correlation with bound cysteine. Therefore, any error in predicting the bound state of cysteines should affect only a small fraction of proteins. When we tested our algorithm on a set of proteins with cysteines in mixed oxidation states the predictive power was sustained.

A limitation of our algorithm is that if more than one fully conserved disulfide bond exists, we cannot predict all disulfide bonds of a protein unambiguously. We demonstrated that the recent expansion of sequence databases made our algorithm applicable to more proteins by an average factor of 1.5 since 1999, which suggests an increasing and wide applicability of this approach in the future. We also examined whether our approach provides a unique aspect of disulfide connectivity prediction, in comparison to other methods. Therefore, we compared the overlap between true positive predictions on the same test set using both our approach and a method developed by Cheng and colleagues (2006), which is one of the best method that is publicly available. The protein test set was composed of proteins sequences from a recent release of Swiss-Prot SP51 (2007) (Boeckmann *et al.*, 2003), filtered as described earlier. We retained only those proteins that shared <30% sequence identity with any other proteins from the training set that was used to train the neural network method developed by Cheng *et al.* (2006). The resulting test set was composed of 275 proteins with 2–4 disulfide bonds and was new to both methods. Our findings showed that our algorithm predicted 135 proteins, of which 83 were correct and 52 were incorrect predictions (61% accuracy). The method of Cheng *et al.*

correctly predicted 79 out of the 135 proteins (58% accuracy). When we compared the true positive predictions of both methods there were 57 overlapping cases. This indicates that an ideal combination of both methods could provide a maximum accuracy of 78%, suggesting a potential 17–20% increase over the current accuracies of these methods if used in combination.

Meanwhile it is very useful if one is able to accurately predict a subset of disulfide bonds that vary as this information can be used as a source input for meta predictors or as a complement for other indirect experimental studies that introduce cross-links. Disulfide bond prediction for proteins where only a subset of disulfides are unconserved has an important implication as it may suggest a structural/functional feature not shared by all members of a protein family. For instance, the T cell immunoglobulin mucin (TIM) protein family provides a recent and interesting example as these proteins were found to have two unique disulfide bonds on top of the canonical disulfide of the immunoglobulin domain. The two non-canonical disulfide bonds support the scaffold of a unique binding site in TIM proteins (Cao *et al.*, 2007).

While past findings indicated that using multiple sequence alignment significantly increases the accuracies of disulfide bond prediction, it remained unclear as to how multiple sequence alignments support the prediction. In the current study we illustrated that part of the contribution of multiple sequence alignment is the identification of the correlated mutation patterns of the query-bounded cysteines. Future studies should evaluate the contribution of correlated mutation pattern of the sequence environment of bound cysteines to disulfide bond prediction.

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REFERENCES

- Altschul,S.F. *et al.* (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res.*, **25**, 3389–3402.
- Boeckmann,B. *et al.* (2003) The SWISS-PROT protein knowledgebase and its supplement TrEMBL in 2003. *Nucleic Acids Res.*, **31**, 365–370.
- Cao,E. *et al.* (2007) T cell immunoglobulin mucin-3 crystal structure reveals a galectin-9-independent ligand-binding surface. *Immunity*, **26**, 311–321.
- Cheek,S. *et al.* (2006) Structural classification of small, disulfide-rich protein domains. *J. Mol. Biol.*, **359**, 215–237.
- Chen,Y.C. and Hwang,J.K. (2005) Prediction of disulfide connectivity from protein sequences. *Proteins*, **61**, 507–512.
- Chen,Y.C. *et al.* (2004) Prediction of the bonding states of cysteines using the support vector machines based on multiple feature vectors and cysteine state sequences. *Proteins*, **55**, 1036–1042.
- Cheng,J. *et al.* (2006) Large-scale prediction of disulphide bridges using kernel methods, two-dimensional recursive neural networks, and weighted graph matching. *Proteins*, **62**, 617–629.
- Chen,B.J. *et al.* (2006) Disulfide connectivity prediction with 70% accuracy using two-level models. *Proteins*, **64**, 246–252.
- Chuang,C.C. *et al.* (2003) Relationship between protein structures and disulfide-bonding patterns. *Proteins*, **53**, 1–5.
- Dekker,J.P. *et al.* (2004) A perturbation-based method for calculating explicit likelihood of evolutionary co-variance in multiple sequence alignments. *Bioinformatics*, **20**, 1565–1572.
- Fariselli,P. and Casadio,R. (2001) Prediction of disulfide connectivity in proteins. *Bioinformatics*, **17**, 957–964.
- Fariselli,P. *et al.* (2002) A neural network based method for predicting the disulfide connectivity in proteins. In Damiani,E. (ed.) *Knowledge Based Intelligent Information Engineering Systems and Allied Technologies (KES)*. IOS Press, Amsterdam, pp. 464–68.
- Ferre,F. and Clote,P. (2005) Disulfide connectivity prediction using secondary structure information and diresidue frequencies. *Bioinformatics*, **21**, 2336–2346.
- Fiser,A. *et al.* (1992) Different sequence environments of cysteines and half cystines in proteins. Application to predict disulfide forming residues. *FEBS Lett.*, **302**, 117–120.
- Fiser,A. and Simon,I. (2000) Predicting the oxidation state of cysteines by multiple sequence alignment. *Bioinformatics*, **16**, 251–256.
- Fiser,A. and Simon,I. (2002) Predicting redox state of cysteines in proteins. *Methods Enzymol.*, **353**, 10–21.
- Gabow,H.N. (1976) An efficient implementation of edmund's algorithm for maximum weight matching on graph. *J. ACM*, **23**, 221–234.
- Gobel,U. *et al.* (1994) Correlated mutations and residue contacts in proteins. *Proteins*, **18**, 309–317.
- Gupta,A. *et al.* (2004) A classification of disulfide patterns and its relationship to protein structure and function. *Protein Sci.*, **13**, 2045–2058.
- Hamilton,N. *et al.* (2004) Protein contact prediction using patterns of correlation. *Proteins*, **56**, 679–684.
- Harrison,P.M. and Sternberg,M.J. (1994) Analysis and classification of disulphide connectivity in proteins. The entropic effect of cross-linkage. *J. Mol. Biol.*, **244**, 448–463.
- Hogg,P.J. (2003) Disulfide bonds as switches for protein function. *Trends Biochem. Sci.*, **28**, 210–214.
- Kadokura,H. *et al.* (2003) Protein disulfide bond formation in prokaryotes. *Annu. Rev. Biochem.*, **72**, 111–135.
- Konagurthu,A.S. *et al.* (2006) MUSTANG: a multiple structural alignment algorithm. *Proteins*, **64**, 559–574.
- Kreisberg,R. *et al.* (1995) Paired natural cysteine mutation mapping: aid to constraining models of protein tertiary structure. *Protein Sci.*, **4**, 2405–2410.
- Larson,S.M. *et al.* (2000) Analysis of covariation in an SH3 domain sequence alignment: applications in tertiary contact prediction and the design of compensating hydrophobic core substitutions. *J. Mol. Biol.*, **303**, 433–446.
- Li,W. and Godzik,A. (2006) Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics*, **22**, 1658–1659.
- Lu,C.H. *et al.* (2007) Predicting disulfide connectivity patterns. *Proteins*, **67**, 262–270.
- Mallick,P. *et al.* (2002) Genomic evidence that the intracellular proteins of archaeal microbes contain disulfide bonds. *Proc. Natl Acad. Sci. USA*, **99**, 9679–9684.
- Mas,J.M. *et al.* (1998) Protein similarities beyond disulphide bridge topology. *J. Mol. Biol.*, **284**, 541–548.
- Mucchielli-Giorgi,M.H. *et al.* (2002) Predicting the disulfide bonding state of cysteines using protein descriptors. *Proteins*, **46**, 243–249.
- Murzin,A.G. *et al.* (1995) SCOP: a structural classification of proteins database for the investigation of sequences and structures. *J. Mol. Biol.*, **247**, 536–540.
- Muskal,S.M. *et al.* (1990) Prediction of the disulfide-bonding state of cysteine in proteins. *Protein Eng.*, **3**, 667–672.
- Neher,E. (1994) How frequent are correlated changes in families of protein sequences? *Proc. Natl Acad. Sci. USA*, **91**, 98–102.
- Poland,D.C. and Scheraga,H.A. (1965) Statistical mechanics of noncovalent bonds in polyamino acids. VIII. Covalent loops in proteins. *Biopolymers*, **3**, 379–399.
- Rai,B.K. *et al.* (2006) MMM: a sequence-to-structure alignment protocol. *Bioinformatics*, **22**, 2691–2692.
- Shindyalov,I.N. *et al.* (1994) Can three-dimensional contacts in protein structures be predicted by analysis of correlated mutations? *Protein Eng.*, **7**, 349–358.
- Thornton,J.M. (1981) Disulphide bridges in globular proteins. *J. Mol. Biol.*, **151**, 261–287.
- Tsai,C.H. *et al.* (2005) Improving disulfide connectivity prediction with sequential distance between oxidized cysteines. *Bioinformatics*, **21**, 4416–4419.
- Vullo,A. and Frasconi,P. (2004) Disulfide connectivity prediction using recursive neural networks and evolutionary information. *Bioinformatics*, **20**, 653–659.
- Wedemeyer,W.J. *et al.* (2000) Disulfide bonds and protein folding. *Biochemistry*, **39**, 7032.
- Wheeler,D.L. *et al.* (2007) Database resources of the National Center for Biotechnology Information. *Nucleic Acids Res.*, **35**, D5–D12.
- Zhao,E. *et al.* (2005) Cysteine separations profiles on protein sequences infer disulfide connectivity. *Bioinformatics*, **21**, 1415–1420.